

PERSISTENT SCENE CHANGE DETECTION

Something gets left behind — a bag, a bike, a cart. Our job is to say exactly when it counts as an encroachment, and exactly when to stop saying so. This is how, and why it works.

0.8465

F1 SCORE

0.9429

PRECISION

0.8760

RECALL

— THE RULE

ONE OBJECT, ONE WINDOW

An object is only ever masked inside a single window — and getting that window's edges right is the entire game.

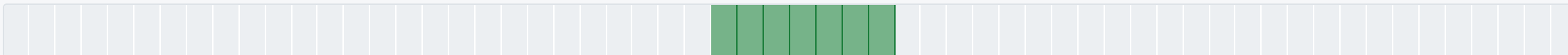


Output is white only inside $[t_0 + P , t_0 + C]$ — never before, never after.

ALMOST EVERY SECOND IS BLANK

One 8-minute video, one 30-second active window. The metric rewards silence far more than it rewards detection.

ONE VIDEO'S TIMELINE — 450 ONE-SECOND SCORING TICKS



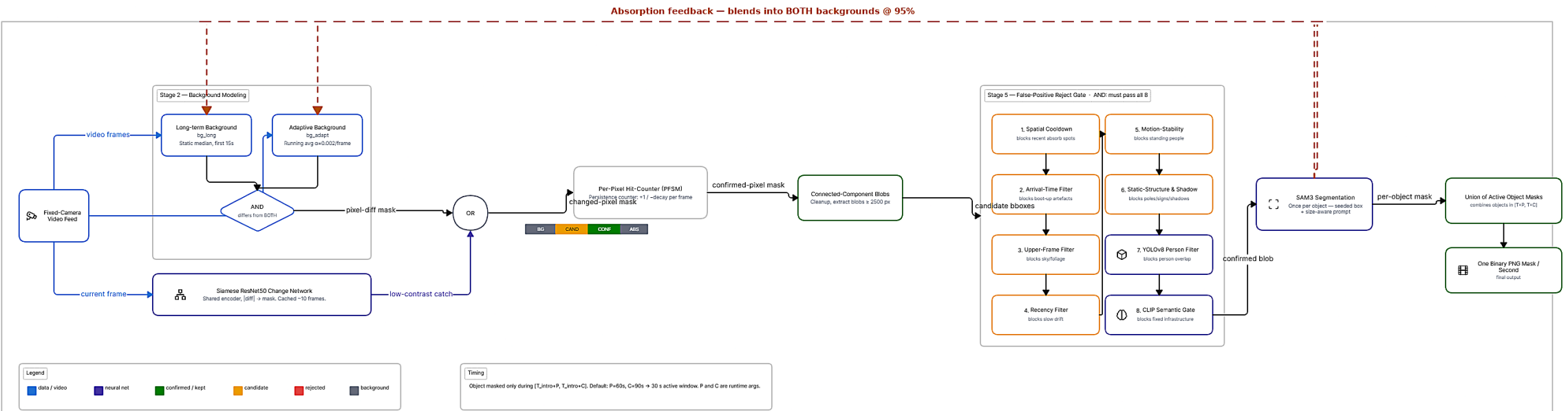
A false alarm on a blank second → $F1 = 0$

Most seconds are blank, so precision — not recall — decides the winner.

Our answer: an aggressive reject stack

Every candidate blob must survive eight independent, mostly training-free filters before it's ever masked.

TWO SIGNALS IN, ONE LOOP CLOSED



TWO MEMORIES, NOT ONE

BG_LONG

Frozen median

First 15 seconds only, guaranteed clean. Never updates — except when an object is formally absorbed.

AND

BG_ADAPT

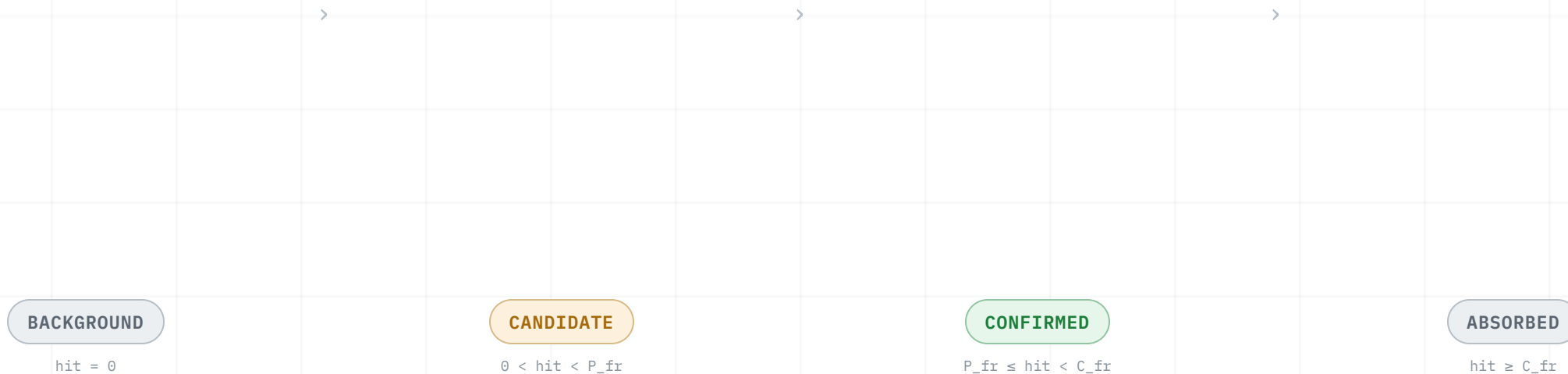
Slow running average

$\alpha = 0.002$ / frame. Tracks gradual drift — sun, shadows — but freezes for any pixel currently confirmed.

A pixel counts as "changed" only if it disagrees with BOTH memories

Illumination drift fools one memory eventually — it never fools both at once. A genuine new object always does.

ONE COUNTER IS THE STATE MACHINE



- **hit += 1** when a pixel differs from both backgrounds this frame; **hit -= decay** otherwise — a brief miss doesn't erase minutes of accumulated persistence.
- No separate state variable exists — thresholding one integer *is* the state machine.

A BLOB MUST SURVIVE EIGHT FILTERS

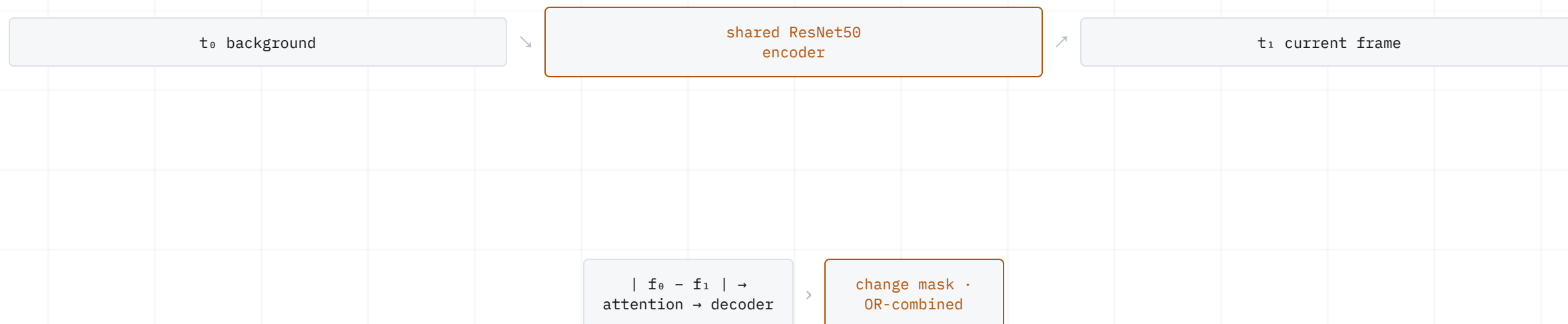
1	Arrival-time	Reject anything confirmed before $\text{bg_window} + 5s + P$ — boot-up artefacts, not real objects.	MUST PASS
2	Upper-frame centroid	Reject blobs centred in the top $\frac{1}{8}$ of frame — sky, foliage, banners.	MUST PASS
3	Recency	$\geq 40\%$ of the blob's pixels must be newly confirmed, not slowly accumulated.	MUST PASS
4	Motion-stability	$< 5\text{px}$ frame-to-frame change inside the blob — standing people still shift 3–6px.	MUST PASS
5	Static-structure & shadow	3 checks: thin/scattered geometry, matches-the-clean-scene texture, cast-shadow signature.	MUST PASS
6	YOLOv8 person	Reject if a detected person's bbox covers $\geq 55\%$ of the blob area.	MUST PASS
7	CLIP semantic gate	Reject if "fixed infrastructure" prompts beat "encroachment object" prompts — reject-only.	MUST PASS
8	Spatial cooldown	No re-registration within C seconds of a nearby absorption — stops re-detecting the same object.	MUST PASS

SEGMENT ONCE, REUSE THE WHOLE WINDOW



- **Seed point** = median of actual persistent pixels (PFSM hit-map), not the naive bbox centre — better for off-centre or irregular shapes.
- **Fixed 100px margin** — enough context for a motorcycle or cart, not so much it pulls a nearby person into a bag's mask.
- **Size-aware text prompt** — "vehicle motorcycle cart" above 3% of frame area, "abandoned object bag luggage" below it.
- Runs exactly **once** per object, at registration — the mask is cached and reused every second of the active window.

A SECOND OPINION FOR LOW-CONTRAST OBJECTS



Tried as the primary signal first — it failed

Trained on outdoor street pairs (VL-CMU-CD + PSCD); run alone on surveillance video it produced 100k+ false "changed" pixels per frame. Demoted to a secondary, OR-combined signal — it can only add detections the pixel-diff missed, never remove ones it found.

— THE ONE BUG THAT MATTERED MOST

ONE IF-STATEMENT

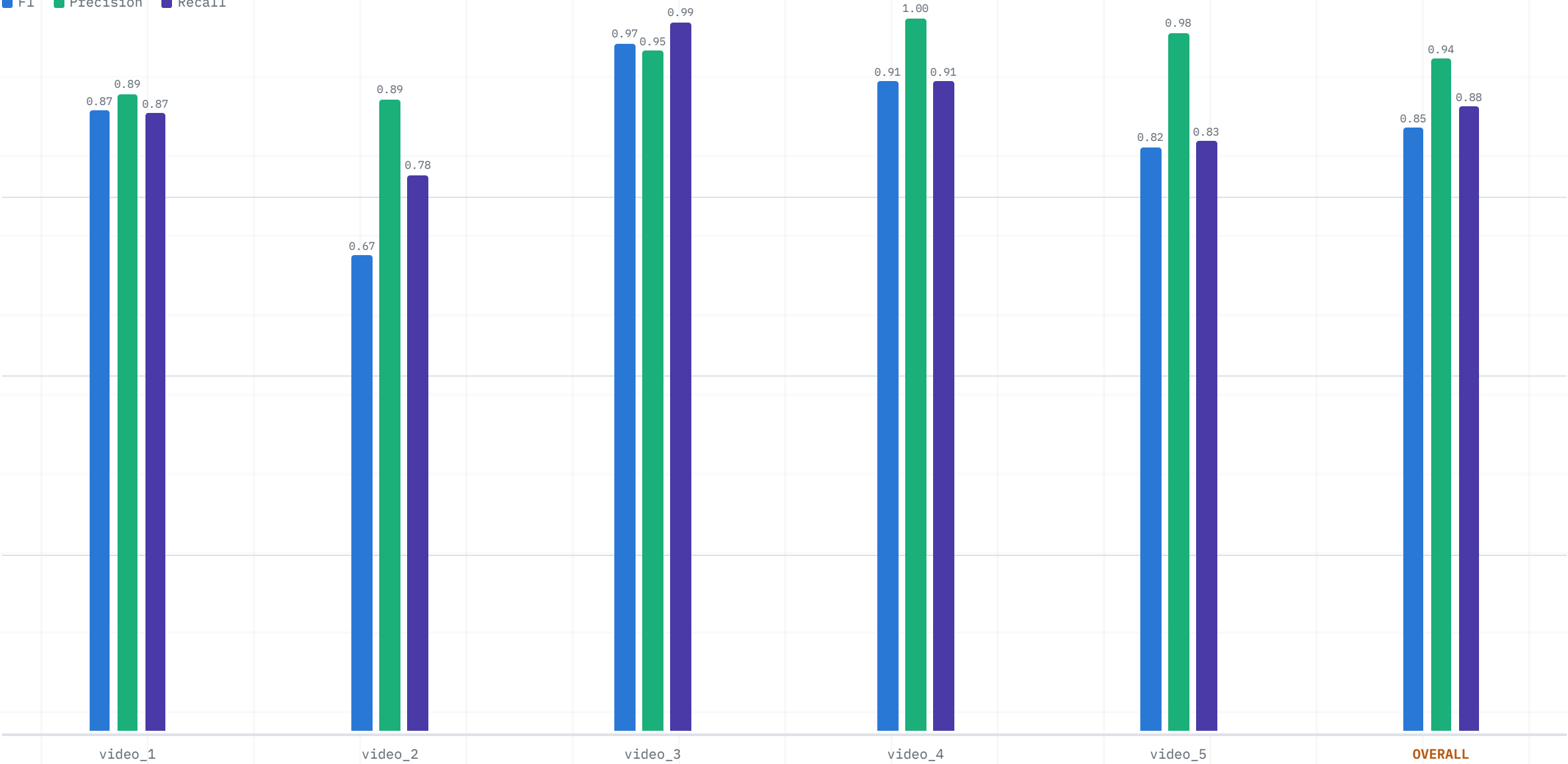
0.06 → 0.85

BLANK-BLANK SCORES 0 BLANK-BLANK SCORES 1

Predicting nothing, correctly, is a *perfect* answer — not a zero. Since most seconds in every video are blank, this single zero-division convention dominated the entire aggregate score. Same predictions, same pixels — one `if` statement changed the measured result by 14×.

FIVE VIDEOS, KNOWN GROUND TRUTH

F1 Precision Recall



WHERE IT STILL SLIPS

Heavy occlusion delays detection

When an object merges into a dense crowd (video_2), the persistence counter can't cleanly separate it — confirmation arrives late.

A few static signboards / poles still trigger

Removing them completely required thresholds that also removed real objects (chair, debris). Kept as the safer trade-off — F1 punishes missed true positives just as hard as false alarms.

— SUMMARY

FIND THE SIGNATURE. REJECT EVERYTHING ELSE.

Appears. Stays. Settles. A dual-background persistence counter finds that signature with an explicit lifecycle tied to p and c; an eight-stage, mostly training-free filter stack keeps precision high against everything that isn't one.

0.8465

F1 · SAMPLE SET

0.9429

PRECISION

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